EQUIDIMENSIONALITY OF RELATIVE CYCLES SATISFIES FPPF DESCENT

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- Functoriality of equidimensionality of relative cycles under base change (relative-cycles chapter, Lemma lemma-equidimensional-functoriality, canonical tag 0H5K). cited at proof lines 1610-1611. Proves the forward direction: base changes of an equidimensional relative cycle are equidimensional.
- Support behaviour of families of cycles under base change (relative-cycles chapter, Lemma lemma-support-family, canonical tag 0H4V). cited at proof lines 1616-1617. Gives Supp(alpha_i) = f_i^{-1}(Supp(alpha)), so the closure of the pulled back support is f_i^{-1}(W).
- Dimension of fibres after base change (Morphisms, Lemma morphisms-lemma-dimension-fibre-after-base-change, canonical tag 02FY). cited at proof lines 1621-1622. With the joint surjectivity of the covering, transfers relative dimension <= r from W_i over S_i back to W over S.

Source reference. Original lemma 0H5K: sources/stacks-chow/tags.txt, source lines 21390–21390. Cited at proof lines 1610-1611: forward direction.

Source reference. Original lemma 0H4V: sources/stacks-chow/tags.txt, source lines 21365–21365. Cited at proof lines 1616-1617: Supp(alpha_i) = f_i^{-1}(Supp(alpha)).

Source reference. Original lemma 02FY: sources/stacks-chow/tags.txt, source lines 3209–3209. Cited at proof lines 1621-1622: transfers relative dimension <= r back to W over S with joint surjectivity.

Lemma 0.2. *Let $f : X \to S$ be a morphism of schemes. Assume S locally Noetherian and f locally of finite type. Let $r \geq 0$ be an integer. Let α be a relative r-cycle on X/S. Let $\{g_i : S_i \to S\}$ be an fppf covering. Then α is equidimensional if and only if each base change $g_i^*\alpha$ is equidimensional.*

Proof. If α is equidimensional, then each $g_i^*\alpha$ is too by Lemma 0H5K. Assume each $g_i^*\alpha$ is equidimensional. Denote W the closure of $\mathrm{Supp}(\alpha)$ in X. Since $g_i : S_i \to S$ is universally open (being flat and locally of finite presentation), so is the morphism $f_i : X_i = S_i \times_S X \to X$. Denote $\alpha_i = g_i^*\alpha$. We have $\mathrm{Supp}(\alpha_i) = f_i^{-1}(\mathrm{Supp}(\alpha))$ by Lemma 0H4V. Since f_i is open, we see that $W_i = f_i^{-1}(W)$ is the closure of

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$\mathrm{Supp}(\alpha_i)$. Hence by assumption the morphism $W_i \to S_i$ has relative dimension $\leq r$. By Morphisms, Lemma 02FY (and the fact that the morphisms $S_i \to S$ are jointly surjective) we conclude that $W \to S$ has relative dimension $\leq r$. $\square$